

Name: _____



Converting terminating decimals to fractions

- 1 A decimal with an infinite repeating pattern of digits is called a _____ decimal.
Tick **1** correct answer

- ☐ fractional
- ☐ proper
- ☐ recurring
- ☐ terminating

- 2 Match each decimal given using dot notation to its equivalent decimal. Write the correct letter in each box

a	$0.\dot{7}\dot{8}$
b	$0.78\dot{9}$
c	$0.\dot{7}8\dot{9}$
d	$0.\dot{7}$
e	$0.7\dot{8}\dot{9}$

	0.789789789...
	0.77777777...
	0.789898989...
	0.789999999...
	0.78787878...



3 Select all the fractions which are equivalent to a recurring decimal. Tick **3** correct answers

☐ $\frac{5}{12}$

☐ $\frac{6}{24}$

☐ $\frac{2}{3}$

☐ $\frac{9}{15}$

☐ $\frac{2}{12}$

4 Use short division to write $\frac{11}{15}$ as a decimal. Tick **1** correct answer

☐ 0.73

☐ 0.7333333333

☐ $0.\dot{7}3$

☐ $0.7\dot{3}$



5 Lucas uses his calculator to write $\frac{7}{18}$ as a decimal. His calculator display shows the number 0.3888888889. Lucas says, "This shows my fraction terminates." Is Lucas correct? Explain why. Tick **1** correct answer

☐ Yes; Lucas' number has 10 digits after the decimal point so it terminates.

☐ Yes; the last digit is 9 not 8 so it doesn't repeat. So it must terminate.

☐ No; all decimals recur eventually.

☐ No; the calculator only has a 12 digit display, the last 8 is rounded up to 9.

6 Aisha writes the recurring decimal $0.61\dot{6}$ as the fraction $\frac{\square}{60}$. What number should she write in the square? _____ Fill in the blank

